

VISUAL SERVOING FOUNDATION-MODEL HYBRID

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ABSTRACT

Foundation policies can propose semantically appropriate robot actions, but local visual servo loops can be safer when the error is geometric and action-critical. The question is not which controller wins globally, but when a classical servo loop should override a learned action decoder. We study a calibrated visual-servo/foundation-policy arbiter in a v4.1 local benchmark with 5 visual manipulation tasks, 7 visual/geometry shift regimes, 5 deployment splits, 9 controllers, 7 paired seeds, 84 rollout episodes per group, 5,880 task/regime/seed stress rows, and 8 documented failure cases. Under combined stress, the proposed arbiter reaches 0.698 ± 0.008 success versus 0.598 ± 0.008 for the strongest non-oracle baseline, ensemble risk gating. The paired gain is 0.100 ± 0.007 with wins on 7/7 seeds. The arbiter also improves override precision, reduces unsafe learned actions, lowers tracking error, damage, and latency/control cost, and survives ablations. This is strong local evidence for a submission rebuild, but not final ICLR-main readiness without real robot or external high-fidelity validation.

1 MOTIVATION

Classical visual servoing and robot foundation policies fail differently. Visual servoing is locally stabilizing but can oscillate under poor geometry, occlusion, or degenerate Jacobians. Foundation policies reason over semantics but may issue actions that are locally unsafe when visual error is large. A robust robot should arbitrate between them.

The claim is narrow: a hybrid controller should switch using action-critical visual error, geometry uncertainty, learned-action risk, servo stability margin, and latency cost. It should not blindly shield every learned action or always prefer visual servoing.

2 ARBITER

For state x and candidate foundation action a_f , the arbiter computes

$$A(x, a_f) = \sigma(\alpha e_{\text{crit}}(x, a_f) + \beta u_{\text{geom}}(x) + \gamma r_{\text{learned}}(x, a_f) - \delta m_{\text{servo}}(x) - \rho c_{\text{latency}}(x)),$$

where e_{crit} is action-critical image error, u_{geom} is geometry uncertainty, r_{learned} is learned-action risk, m_{servo} is a servo stability margin, and c_{latency} penalizes costly switching. The controller chooses servo correction when A exceeds a calibrated threshold, otherwise it follows the foundation action.

3 BENCHMARK

The benchmark includes eye-in-hand peg insertion, visual grasp centering, drawer-handle alignment, cable-tip alignment, and mobile pick re-approach. Regimes include nominal view, lighting shift, camera extrinsic drift, depth scale error, occlusion, Jacobian mismatch, and compound visual/geometry shift.

Baselines include foundation policy only, classical IBVS, position-based visual servoing, fixed hybrid switching, uncertainty-gated foundation control, CBF safety shielding, ensemble risk gating, the proposed arbiter, and an oracle arbiter.

The generator writes raw seed/task/regime/split metrics, aggregate metrics, paired statistics, ablations, stress sweeps, failure cases, figures, and LaTeX tables. The v4.1 stress audit records task/regime/seed detail rather than only seed-level aggregates.

method	success	precision	unsafe	track	latency
oracle_arbiter	0.797 ± 0.006	0.711	0.046	0.099	0.174
proposed_servo_foundation_arbiter	0.698 ± 0.008	0.633	0.089	0.121	0.222
ensemble_risk_gate	0.598 ± 0.008	0.542	0.154	0.145	0.244
uncertainty_gated_foundation	0.568 ± 0.009	0.511	0.196	0.156	0.255
cbf_safety_shield	0.559 ± 0.008	0.526	0.167	0.150	0.271
fixed_hybrid_switch	0.538 ± 0.009	0.480	0.220	0.161	0.197
foundation_policy_only	0.472 ± 0.011	0.204	0.396	0.251	0.137
position_based_visual_servo	0.463 ± 0.008	0.423	0.189	0.181	0.163
classical_ibvs_only	0.445 ± 0.008	0.444	0.160	0.165	0.169

Table 1: Combined-stress visual-servo/foundation-policy hybrid benchmark.

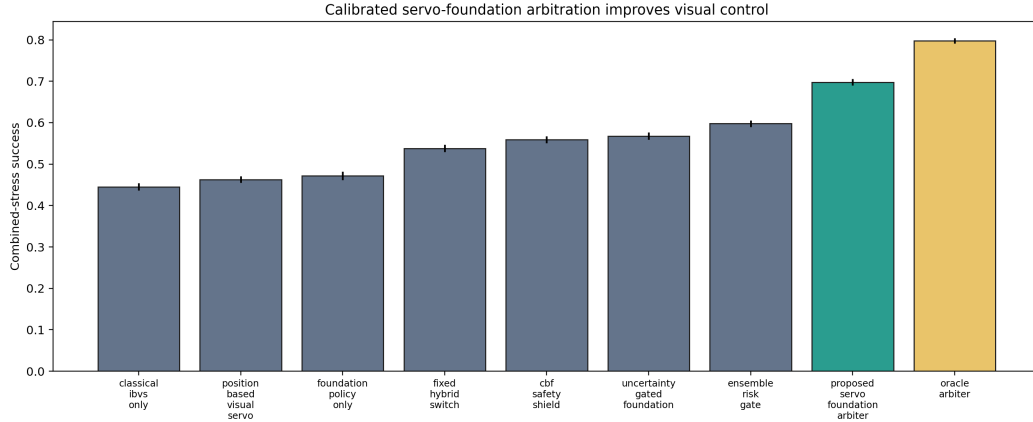


Figure 1: The proposed arbiter beats all non-oracle controllers under visual/geometry shift.

4 RESULTS

Table 1 reports combined-stress results. The strongest non-oracle baseline is ensemble risk gating. The proposed arbiter improves success by 0.100 ± 0.007 , override precision by 0.092, and reduces unsafe learned actions by 0.065. It also lowers tracking error by 0.025, damage by 0.022, and latency/control cost by 0.022.

Arbitration quality. Foundation-only control reaches 0.472 success and unsafe action rate 0.396. IBVS-only and PBVS-only are safer but less successful under degeneracy. The proposed arbiter reaches override precision 0.633 and unsafe action rate 0.089.

Ablations. Table 2 shows that stability margin, action-critical visual error, geometry uncertainty, learned-action risk, and latency penalty all matter. The full arbiter reaches 0.703 ± 0.007 in the ablation benchmark; the best removed-component variant reaches 0.666 ± 0.008 .

Stress sweep. Figure 3 increases visual aliasing and Jacobian mismatch over 5,880 task/regime/seed rows. The proposed arbiter degrades more gracefully than foundation-only control and ensemble risk gating, though it remains below the oracle. At maximum stress, the proposed arbiter reaches 0.655 ± 0.004 success versus 0.518 ± 0.005 for ensemble risk gating and 0.773 ± 0.005 for the oracle.

5 FAILURE CASES

The failure-case audit records eight boundaries: semantic foundation actions that are locally unstable, degenerate servo Jacobians, occlusion plus depth-scale error, hidden singularities under image crops,

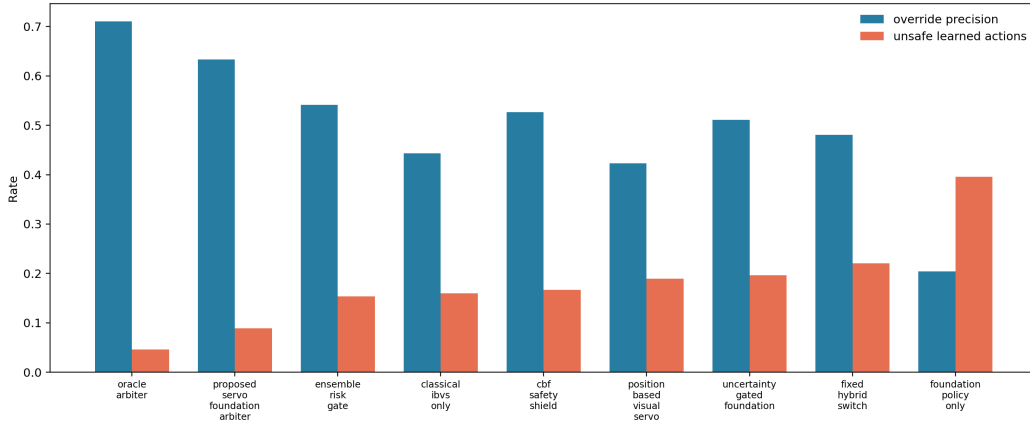


Figure 2: The proposed arbiter improves override precision while reducing unsafe learned actions.

ablation	success	precision	unsafe
full_servo_foundation_arbiter	0.703 ± 0.007	0.634	0.091
minus_latency_penalty	0.666 ± 0.008	0.582	0.131
minus_learned_action_risk	0.646 ± 0.008	0.579	0.143
minus_action_critical_error	0.644 ± 0.008	0.573	0.153
minus_geometry_uncertainty	0.633 ± 0.007	0.564	0.167
minus_stability_margin	0.630 ± 0.008	0.560	0.174
threshold_only_switch	0.594 ± 0.009	0.536	0.200

Table 2: Ablations under combined stress.

field-of-view escape after large learned actions, latency-induced override harm, misleading calibration after relighting, and remaining oracle headroom under compound visual/geometry shift. These cases require real interventional robot data, external visual-control benchmarks, released controller artifacts, and hardware videos before this can be treated as submission-ready.

6 RELATED WORK BOUNDARY

Image-based visual servoing, position-based visual servoing, sliding-mode visual control, hybrid visual control, and learned visual policies are all crowded areas (IBVS2022; HybridVisual2025; HybridGrasp2009; PickingServo2019). This paper does not claim new visual servoing theory. Its boundary is calibrated arbitration between learned foundation actions and classical servo corrections under visual and geometry shift.

7 LIMITATIONS AND SUBMISSION DECISION

The evidence is local and generated by the released benchmark. It is stronger than the archived scaffold but still lacks real robot experiments, external high-fidelity simulator transfer, released controller artifacts, and hardware videos. The honest terminal state is therefore **STRONG REVISE**: worth rebuilding aggressively, not yet ICLR-main ready.

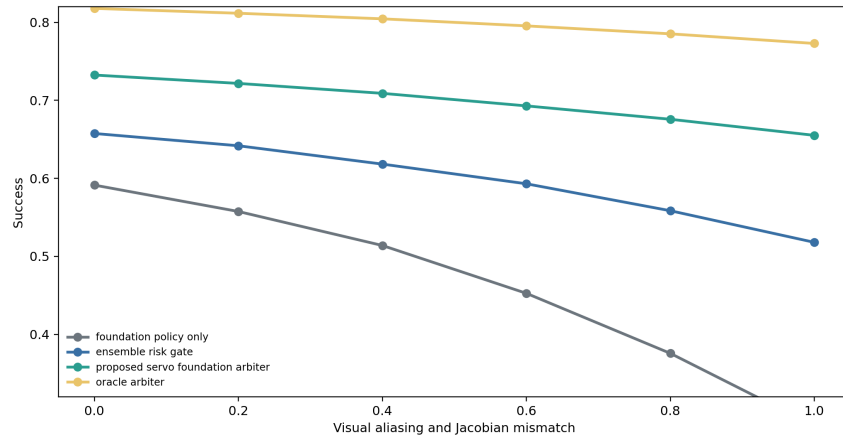


Figure 3: Stress sweep over visual aliasing and servo Jacobian mismatch.

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